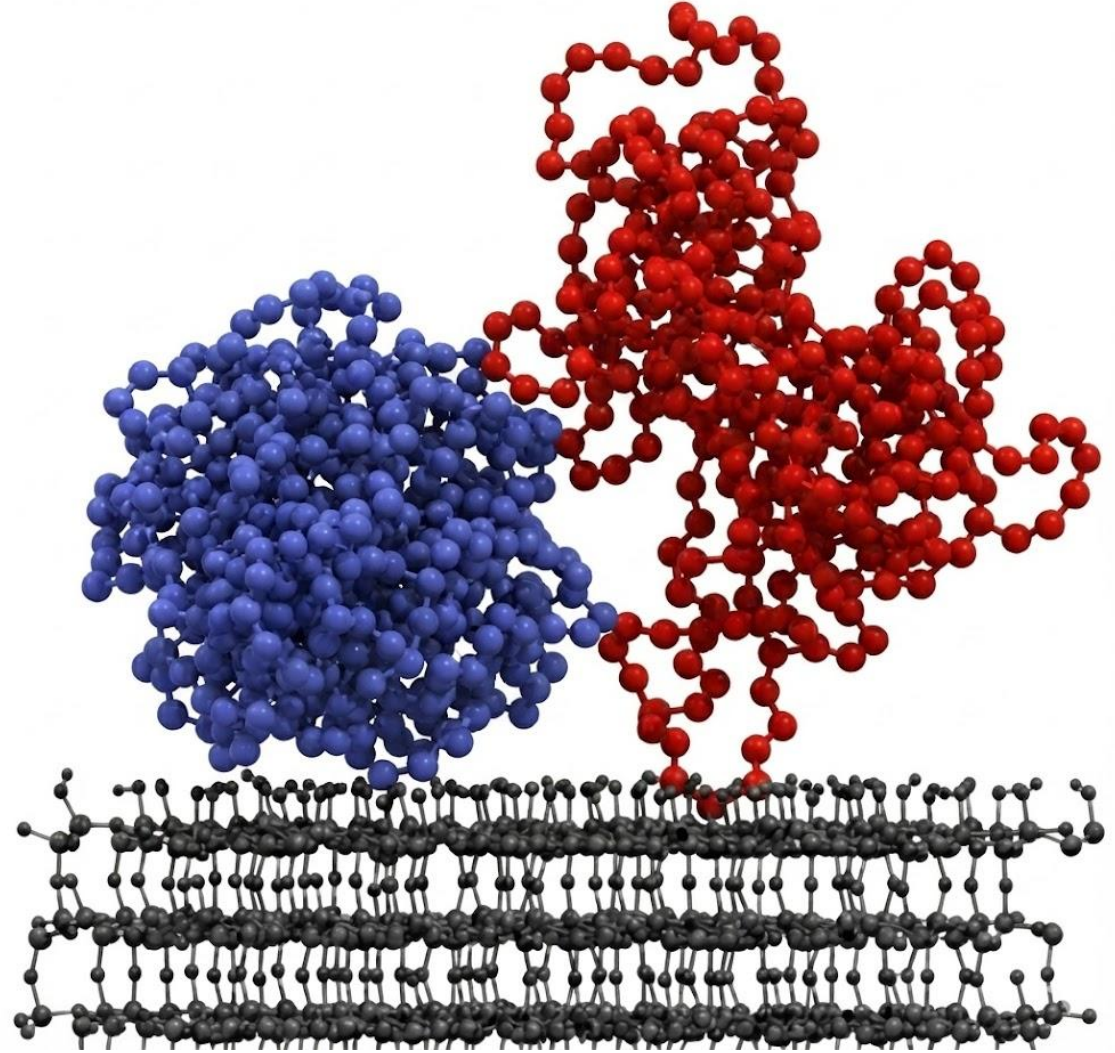
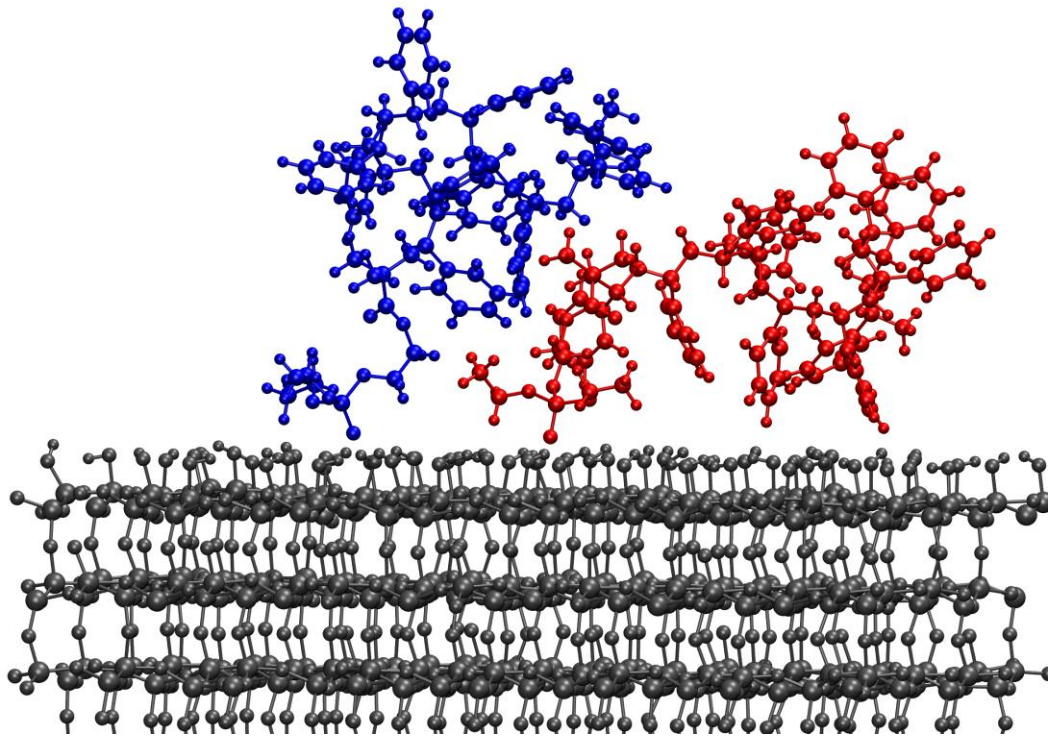
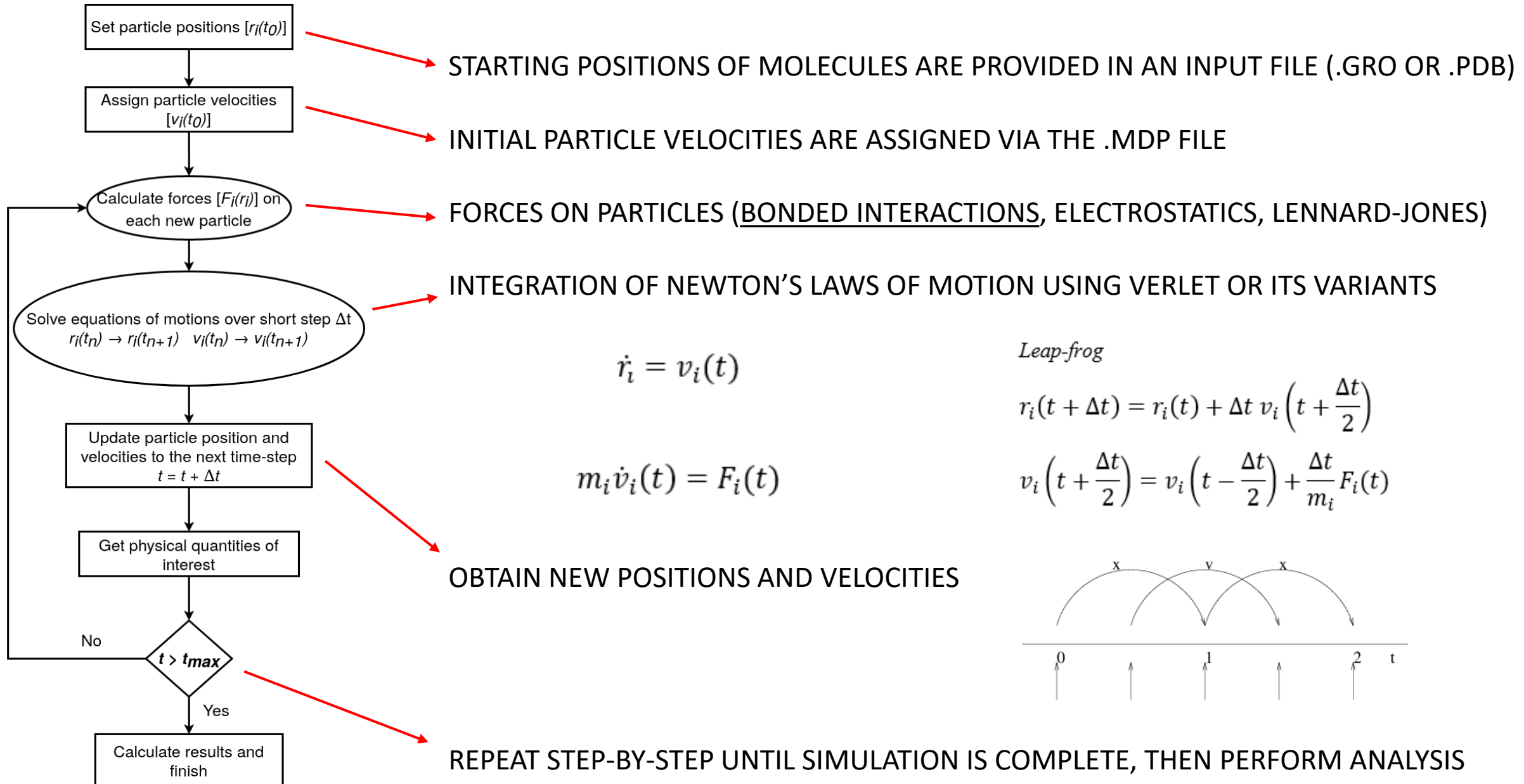


MOLECULAR MODELING: atomistic and coarse-grained models



MOLECULAR MODELING: METHODS



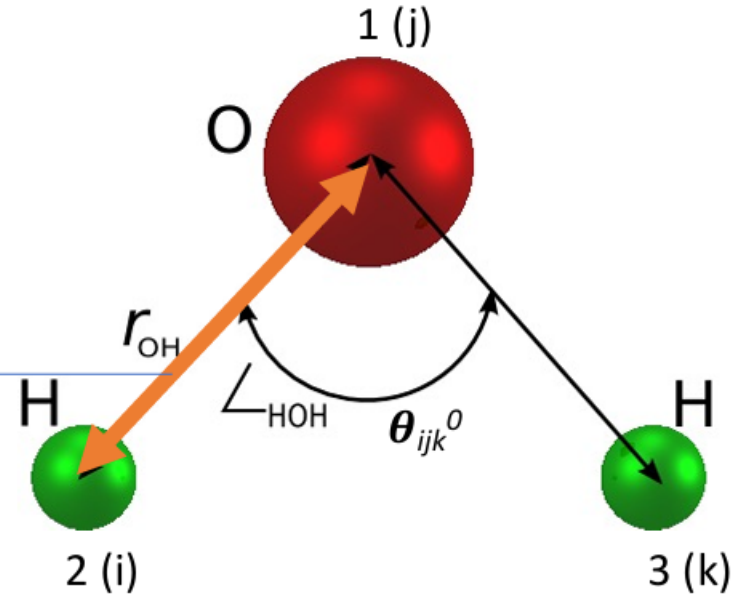
MOLECULAR MODELING: BONDED INTERACTIONS

Bonded Interactions

Bond Stretching

$$V_b(r_{ij}) = \frac{1}{2}k_{ij}^b(r_{ij} - b_{ij})^2$$

$$\mathbf{F}_i(\mathbf{r}_{ij}) = k_{ij}^b(r_{ij} - b_{ij})\frac{\mathbf{r}_{ij}}{r_{ij}}$$



Bending

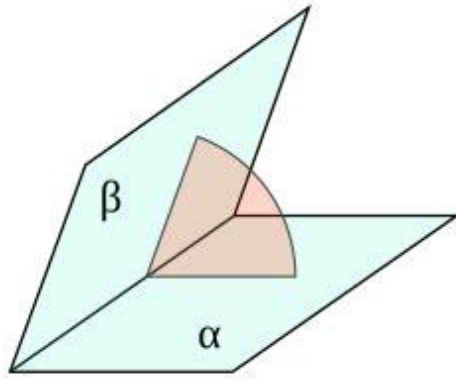
$$V_a(\theta_{ijk}) = \frac{1}{2}k_{ijk}^\theta(\theta_{ijk} - \theta_{ijk}^0)^2$$

$$V_a(\theta_{ijk}) = \frac{1}{2}k_{ijk}^\theta(\cos(\theta_{ijk}) - \cos(\theta_{ijk}^0))^2$$

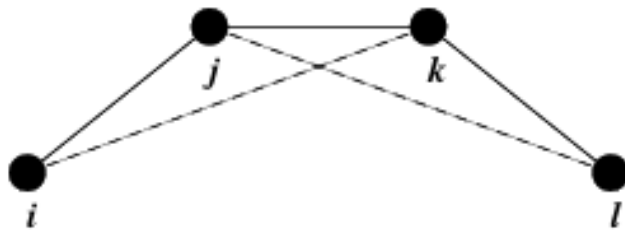
Il calcolo dell'energia di atomi e/o molecole si avvale di funzioni e parametri. L'insieme di queste funzioni e parametri prende il nome di **Force field** o *campo di forze*,

Pag. 411-417 gromacs handbook 2025.1
'5.5.2: BONDED INTERACTIONS'

MOLECULAR MODELING: BONDED INTERACTIONS



Proper dihedrals: Torsions

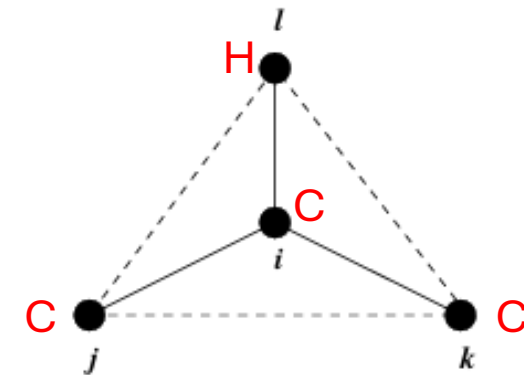


$$V_d(\phi_{ijkl}) = k_\phi (1 + \cos(n\phi - \phi_s))$$

$$V_{rb}(\phi_{ijkl}) = \sum_{n=0}^5 C_n (\cos(\psi))^n$$

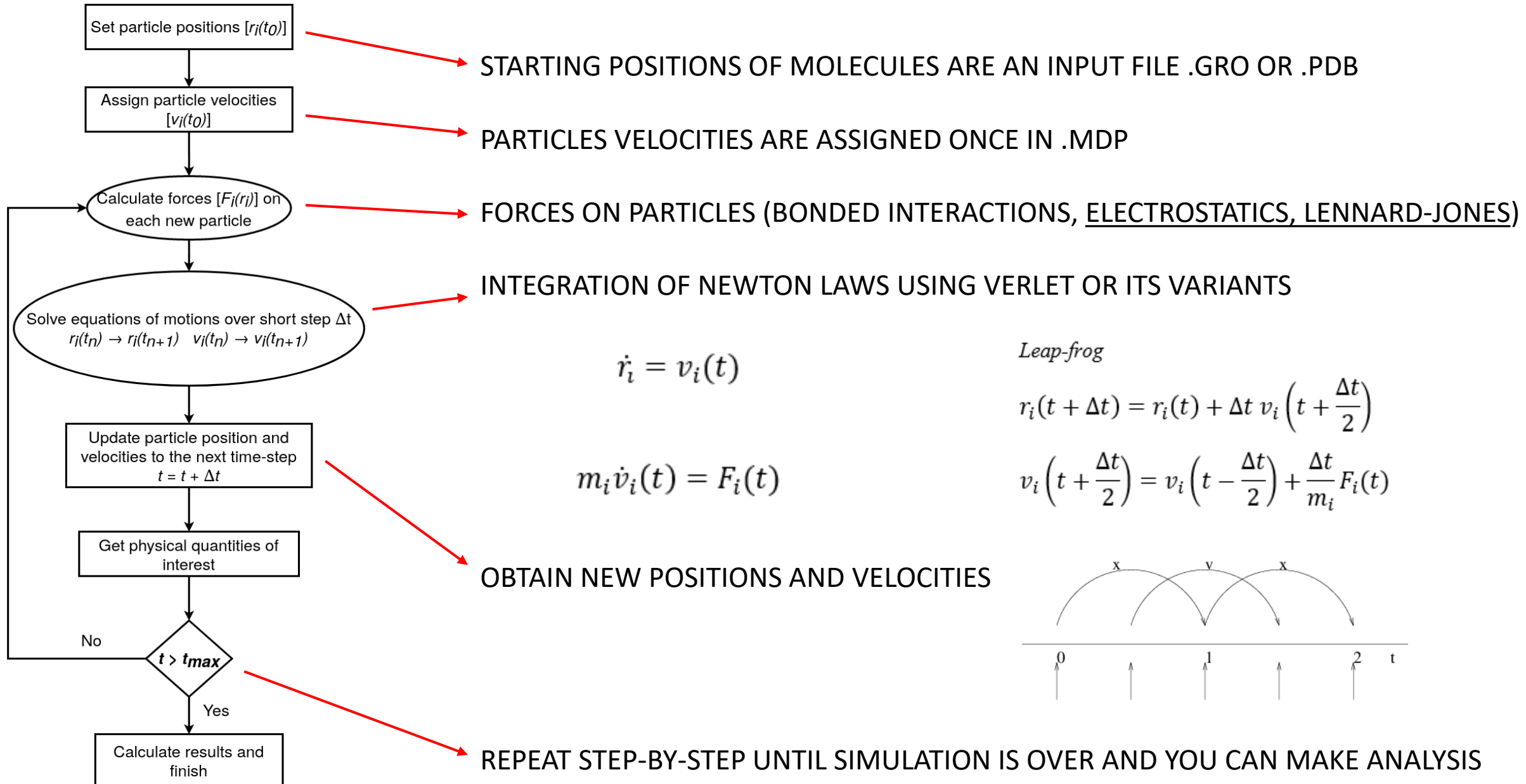
Pag. 417-422 & 470
gromacs handbook 2025.1
'5.5.2: BONDED INTERACTIONS'

Improper dihedrals: Chirality and aromatics



$$V_{id}(\xi_{ijkl}) = \frac{1}{2} k_\xi (\xi_{ijkl} - \xi_0)^2$$

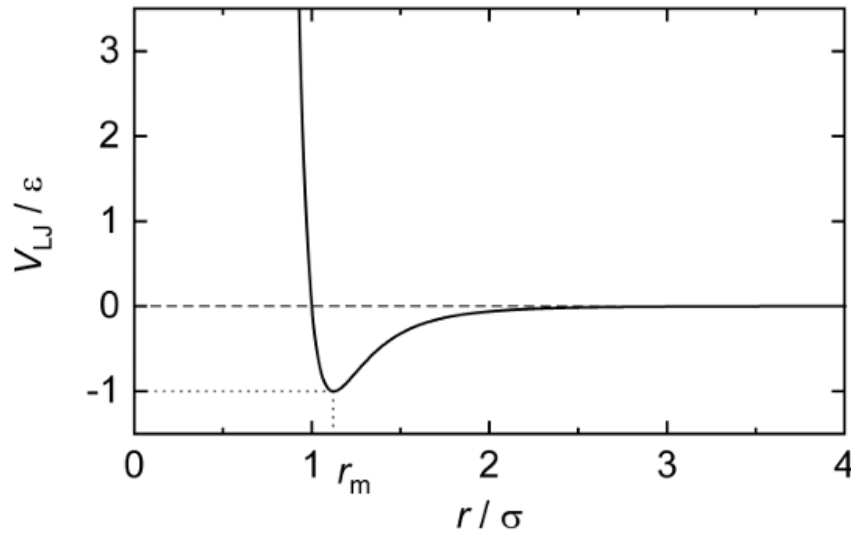
MOLECULAR MODELING: METHODS



MOLECULAR MODELING: NON-BONDED INTERACTIONS

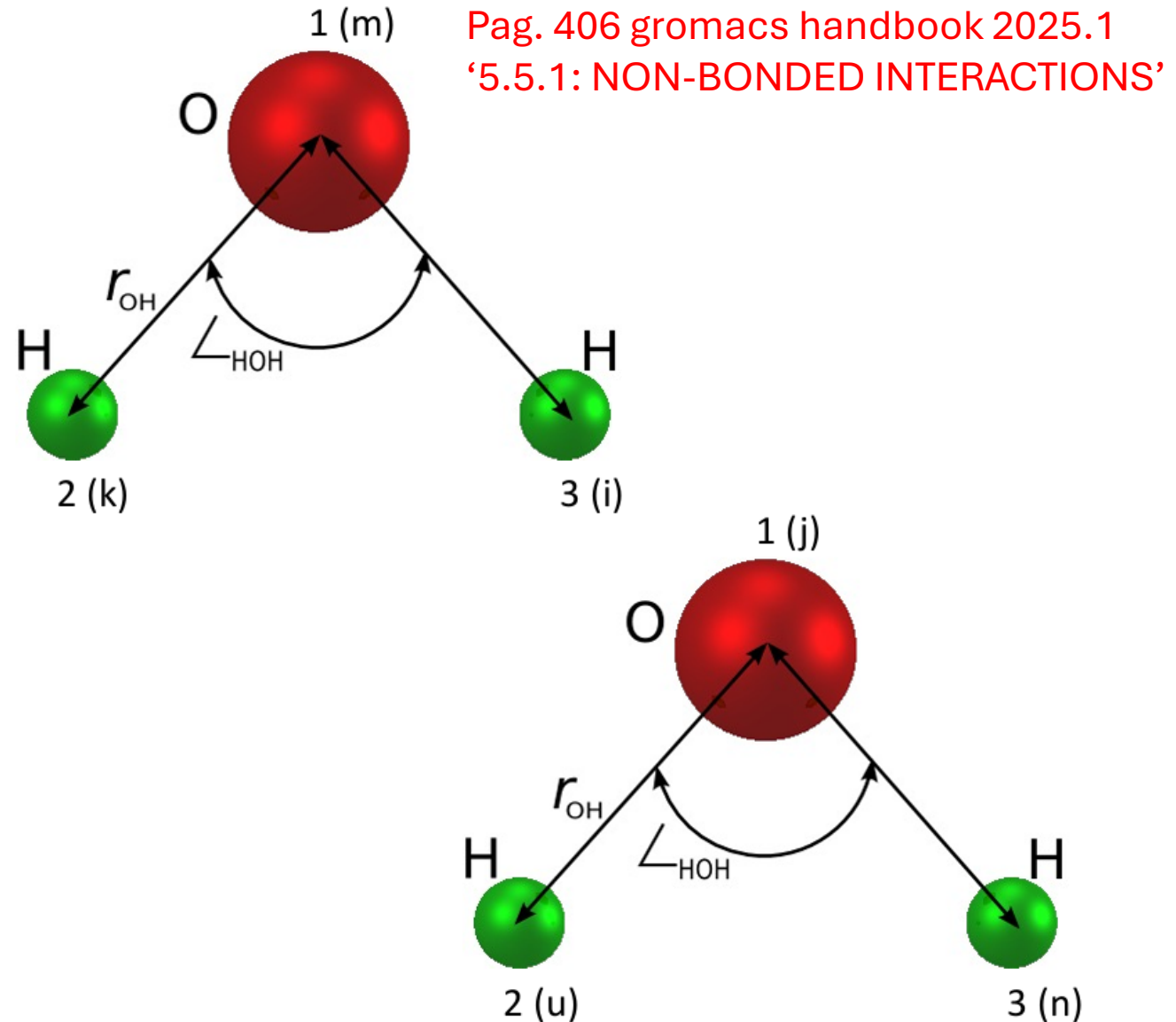
Lennard-Jones Interactions

$$\phi_{LJ}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right]$$

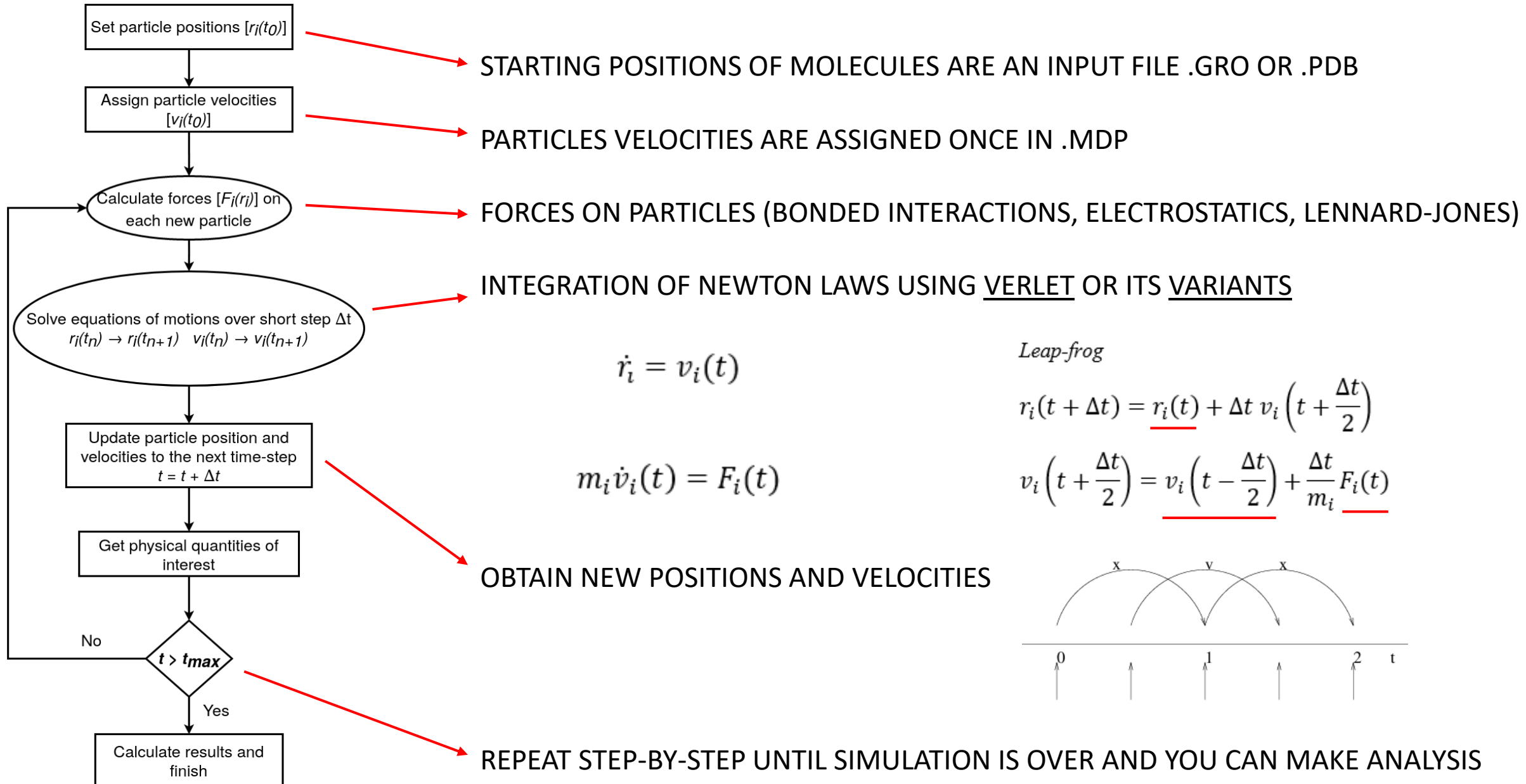


Electrostatic Interactions

$$v^{qq}(r_{ij}) = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$$



MOLECULAR MODELING: METHODS



MOLECULAR MODELING: NECESSARY DATA

SIMULATION SETTINGS
(.mdp)



- SIMULATION DURATION
- TEMPERATURE AND PRESSURE
- ALGORITHMS
- MANY OTHERS...

COORDINATES
(.gro OR .pdb)



- STARTING POSITIONS OF MOLECULES IN THE SYSTEM

TOPOLOGY
(.top AND .itp)



- STRUCTURE ATOMS
- BONDED INTERACTIONS
- NON BONDED INTERACTIONS

MOLECULAR MODELING: SETTINGS(.mdp)

```
; RUN CONTROL PARAMETERS
integrator          = md
; Start time and timestep in ps
tinit              = 0.00
dt                 = 0.002 ;(2 femtosecondi)
nsteps             = 50000 ;(la simulazione dura 100 ps)
; For exact run continuation or redoing part of a run
init_step          = 0
; Part index is updated automatically on checkpointing (keeps files separate)
simulation-part     = 1
; mode for center of mass motion removal
comm-mode          = Linear
; number of steps for center of mass motion removal
nstcomm           = 250
; group(s) for center of mass motion removal
comm-grps          =
```

```
; OPTIONS FOR WEAK COUPLING ALGORITHMS
; Temperature coupling
tcoupl             = v-rescale
nsttcouple         = -1
nh-chain-length    = 10
print-nose-hoover-chain-variables = no
; Groups to couple separately
tc-grps            = system
; Time constant (ps) and reference temperature (K)
tau_t              = 0.1
ref_t              = 298
; pressure coupling
Pcoupl             = c-rescale
Pcoupltype         = Isotropic
nstpcouple         = -1
; Time constant (ps), compressibility (1/bar) and reference P (bar)
tau-p              = 1
compressibility     = 4.5e-5
ref-p              = 1.0
; Scaling of reference coordinates, No, All or COM
refcoord-scaling   = No
```

```
; GENERATE VELOCITIES FOR STARTUP RUN
gen-vel            = yes
gen-temp           = 298
gen-seed           = 19000433251425
```

```
; NEIGHBORSEARCHING PARAMETERS
; cut-off scheme (Verlet: particle based cut-offs, group: using charge groups)
cutoff-scheme      = Verlet
; nblist update frequency
nstlist            = 10
; ns algorithm (simple or grid)
ns_type            = grid
; Periodic boundary conditions: xyz, no, xy
pbc                = xyz
periodic-molecules = no
; Allowed energy error due to the Verlet buffer in kJ/mol/ps per atom,
; a value of -1 means: use rlist
verlet-buffer-tolerance = 0.005
; nblist cut-off
rlist              = 1.0
; long-range cut-off for switched potentials
rlistlong          = -1
nstcalclr          = -1

; OPTIONS FOR ELECTROSTATICS AND VDW
; Method for doing electrostatics
coulombtype        = PME
coulomb-modifier    = Potential-shift-Verlet
rcoulomb-switch    = 1.0
rcoulomb            = 1.0
; Relative dielectric constant for the medium and the reaction field
epsilon-r           = 1.0
epsilon-rf          = 79.2
; Method for doing Van der Waals
vdw-type           = Cut-off
vdw-modifier        = Potential-shift-Verlet
; cut-off lengths
rvdw-switch        = 1.0
rvdw                = 1.0
; Apply long range dispersion corrections for Energy and Pressure
DispCorr           = EnerPres
; Extension of the potential lookup tables beyond the cut-off
table-extension    = 1
; Separate tables between energy group pairs
energygrp-table     =
; Spacing for the PME/PPPM FFT grid
fourierspacing     = 0.16
```

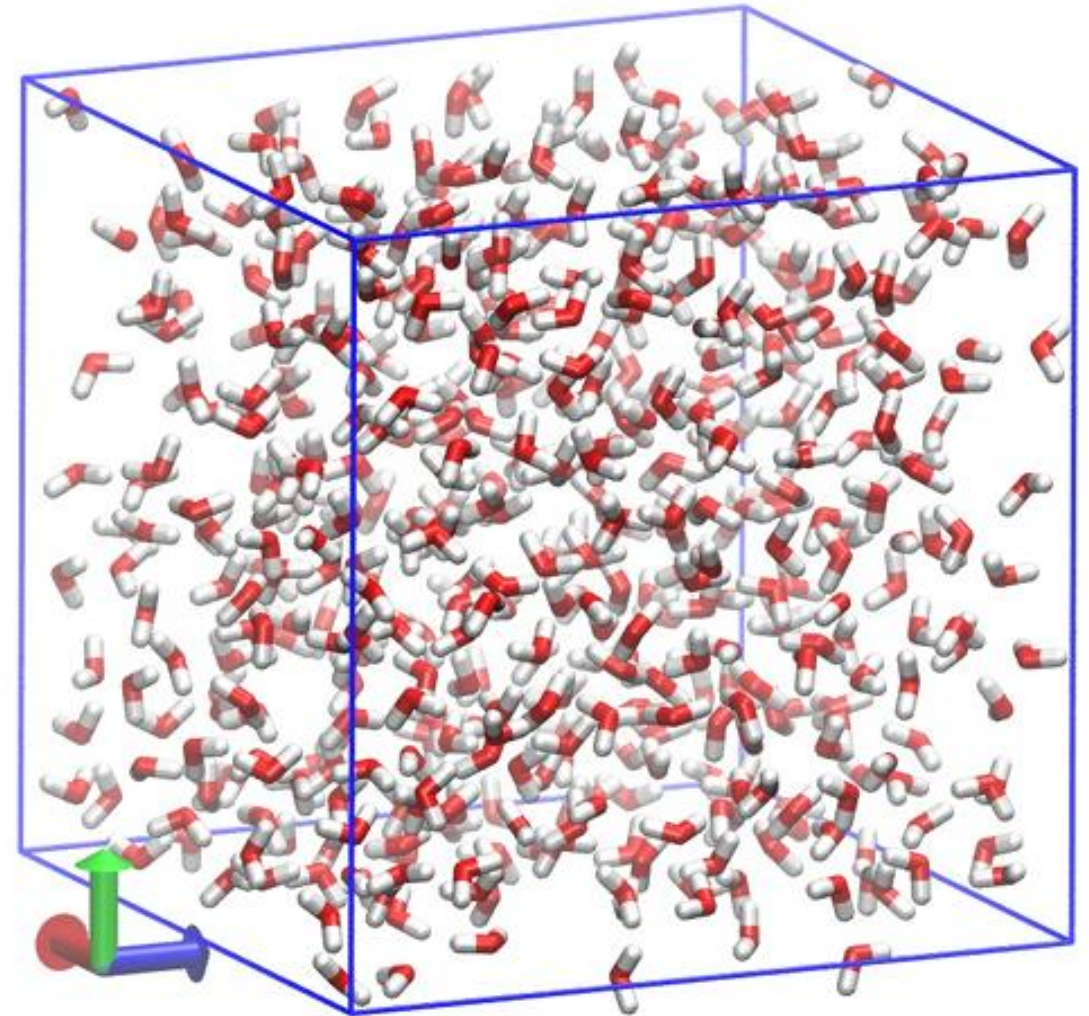
MOLECULAR MODELING: COORDINATES (.gro or pdb)

H2O

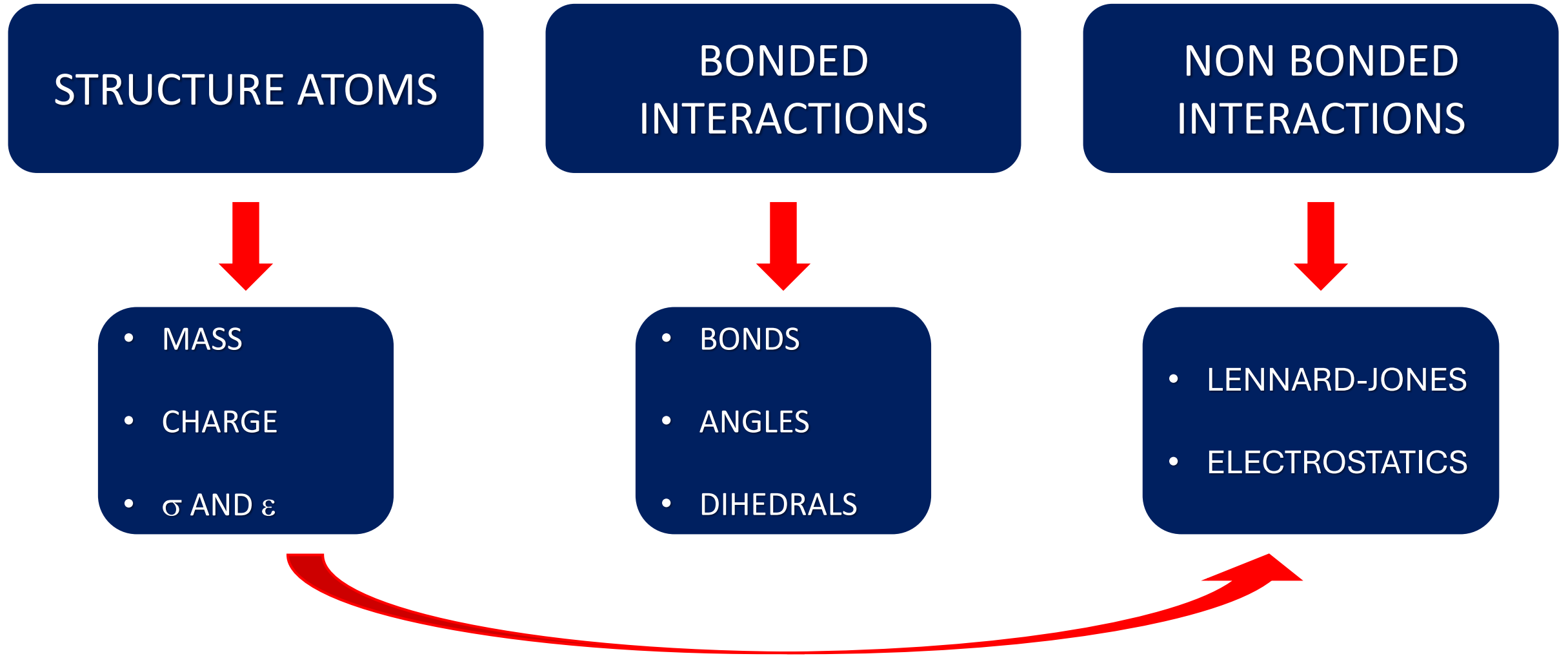
600

1H2O	OW	1	3.971	3.852	4.679
1H2O	HW	2	4.066	3.852	4.688
1H2O	HW	3	3.940	3.943	4.678
2H2O	OW	4	0.787	4.994	4.656
2H2O	HW	5	0.874	5.025	4.688
2H2O	HW	6	0.751	4.935	4.723
3H2O	OW	7	3.317	0.807	1.740
3H2O	HW	8	3.356	0.839	1.824
3H2O	HW	9	3.236	0.856	1.726
4H2O	OW	10	3.294	0.813	0.312
4H2O	HW	11	3.297	0.889	0.369
4H2O	HW	12	3.226	0.753	0.347
5H2O	OW	13	3.503	0.926	5.374
5H2O	HW	14	3.485	0.870	5.453
5H2O	HW	15	3.418	0.957	5.342

199H2O	OW	595	2.868	1.747	0.405
199H2O	HW	596	2.838	1.719	0.319
199H2O	HW	597	2.945	1.804	0.395
200H2O	OW	598	4.975	0.569	2.625
200H2O	HW	599	5.034	0.609	2.561
200H2O	HW	600	4.956	0.478	2.593
5.50000	5.50000	5.50000			



MOLECULAR MODELING: TOPOLOGY



MOLECULAR MODELING: NECESSARY DATA (.itp)

```
;#include "oplsaa.ff/forcefield.itp"
```

```
[ defaults ]
```

```
; nbfunc      comb-rule    gen-pairs    fudgeLJ    fudgeQQ  
  1             3           yes           0.5        0.5
```

```
[ atomtypes ]
```

```
; name      mass      charge      ptype      sigma(nm)      epsilon(kJ/mol)  
  OW       15.9994      0.0         A        3.16557e-01     6.50194e-01  
  HW        1.00800      0.0         A        0.00000e+00     0.00000e+00
```

```
[ bondtypes ]
```

```
; i      j      func      b0(nm)      k(kJ/mol·nm²)  
  OW     HW      1         0.1          345000
```

```
[ angletypes ]
```

```
; i      j      k      func       $\theta$  (°)      k $\theta$  (kJ/mol·rad²)  
  HW     OW     HW      1         109.470        383
```

```
[ moleculetype ]
```

```
; Name      nrexcl  
  H2O         2
```

```
[ atoms ]
```

```
; nr      type      resnr      residue      atom      cgnr      charge      mass  
  1        OW         1         H2O          OW         1        -0.8476      15.9994  
  2        HW         1         H2O          HW         1         0.4238       1.00800  
  3        HW         1         H2O          HW         1         0.4238       1.00800
```

```
[ bonds ]
```

```
; i      j      func  
  1        2         1  
  1        3         1
```

```
[ angles ]
```

```
; i      j      k      func  
  2        1        3         1
```

Flexible water model: topolwater.itp

MOLECULAR MODELING: NECESSARY DATA (.itp)

```
#include "../oplsaa.ff/forcefield.itp"

[ moleculetype ]
; molname      nrexcl
  H2O           2

[ atoms ]
;  nr  type  resnr  residue  atom  cgnr  charge  mass
   1  op1s_116  1    H2O     OW     1    -0.8476
   2  op1s_117  1    H2O     HW     1     0.4238
   3  op1s_117  1    H2O     HW     1     0.4238

#ifdef FLEXIBLE
[ settles ]
; OW  funct  doh  dhh
1     1     0.1  0.16330
#else
[ exclusions ]
1     2     3
2     1     3
3     1     2
#else
[ bonds ]
; i  j  funct  length  force.c.
1   2   1     0.1    345000  0.1    345000
1   3   1     0.1    345000  0.1    345000

[ angles ]
; i  j  k  funct  angle  force.c.
2   1  3   1    109.47  383    109.47  383
#endif
```

Rigid water model: spce.itp

MOLECULAR MODELING: NECESSARY DATA (.top)

```
;
;      File 'topol.top' was generated
;      By user: paolo
;      At date: 06/02/2025
;
;      This is your topology file
;
; Include forcefield parameters

#include "spce.itp"
[ system ]
; Name
H2O

[ molecules ]
; Compound      #mols
H2O              200
~
```

topol.top

MOLECULAR MODELING: SIMULATION

‘Packing’ input files in simulation starter file (.tpr)

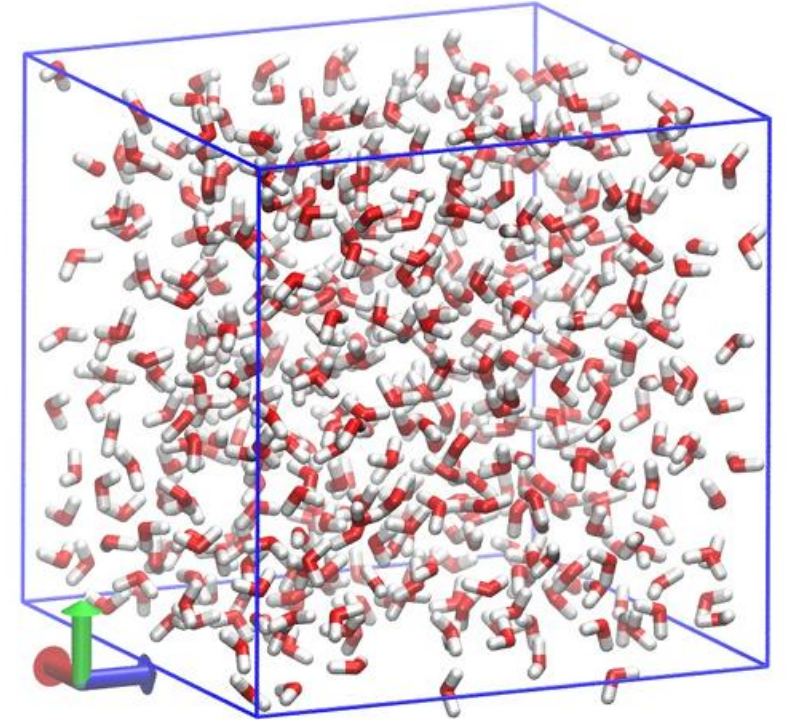
➤ `gmx grompp -f mdprod.mdp -p topol.top -c coord.gro -o simul.tpr`

How to run a simulation

➤ `gmx mdrun -s simul.tpr`

Output files

- traj.trr
- ener.edr
- md.log
- confout.gro
- state.cpt and state_prev.cpt
- mdout.mdp



MOLECULAR MODELING: OUTPUT FILES

traj.trr: total trajectory of molecules

- `gmx trjconv -f traj.trr -s simul.tpr -o traj.gro -pbc mol`
- Select «Group 0 (System)»

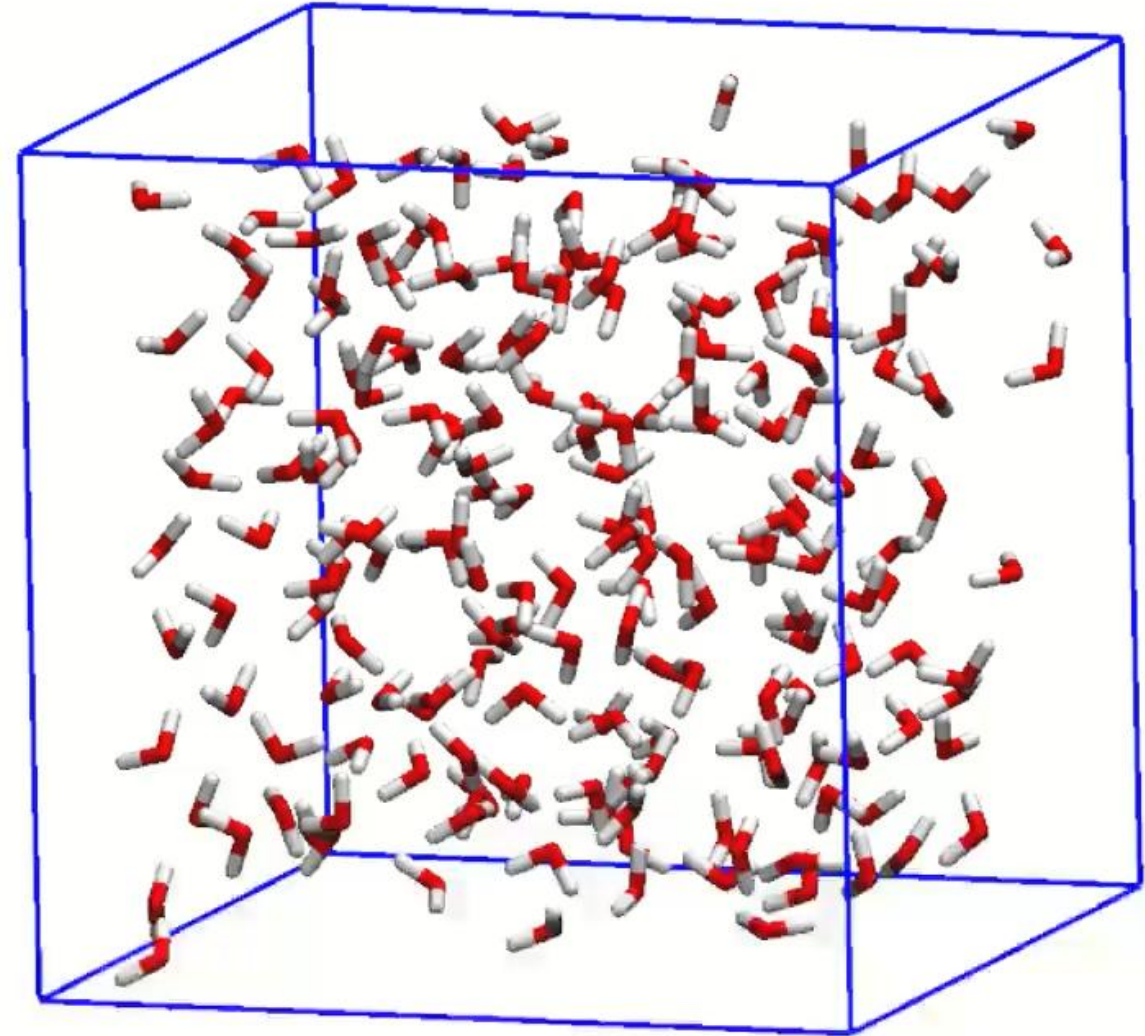
Open it on vmd to see how the molecules have moved!

TIPS FOR VMD:

Type «draw pbcbox» in the VMD window to draw the simulation box

Go to Display and select «Orthographic» and «Depth Cueing» for a better visualization

Play with Representation and Colors

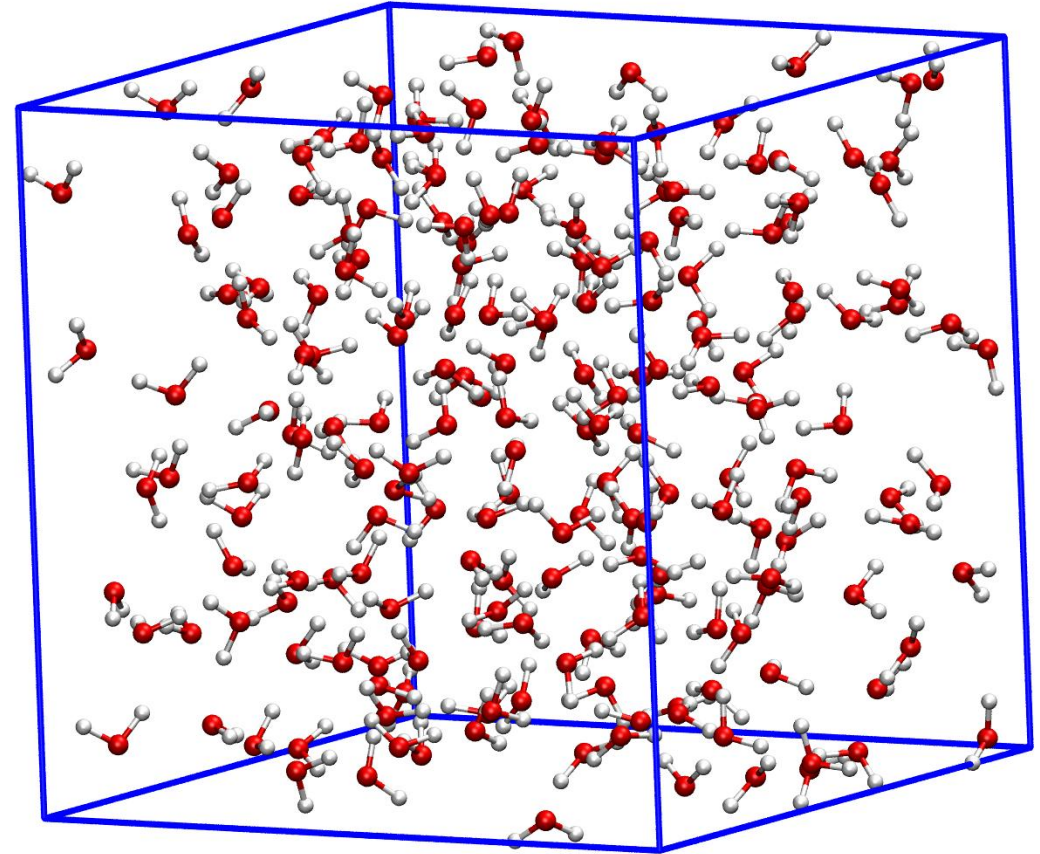


MOLECULAR MODELING: OUTPUT FILES

confout.gro: last frame of trajectory

Open it on vmd and let's save a high quality image:

- Change background color and molecules representation as you prefer
- Zoom in and zoom out and change perspective as you prefer
- File > Render > Tachyon.
- Then «browse» and select where to save the image. Give it a name without format!!!
- Change the resolution after «BMP» (es: -res 2400 1800)
- Start Rendering



Confout.gro contains the output coordinates which can be the input coordinates for a subsequent simulation...

MOLECULAR MODELING: OUTPUT FILES

md.log: a ‘Chronology’ of the simulation

```
Command line:
  gmx mdrun -s topol.tpr

GROMACS version: 2024.4
Precision: mixed
Memory model: 64 bit
MPI library: thread_mpi
OpenMP support: enabled (GMX_OPENMP_MAX_THREADS = 128)
GPU support: disabled
SIMD instructions: AVX2_256
CPU FFT library: fftw-3.3.8-sse2-avx-avx2-avx2_128
GPU FFT library: none
Multi-GPU FFT: none
RDTSOP usage: enabled
TNG support: enabled
Hwloc support: disabled
Tracing support: disabled
C compiler: /usr/bin/cc GNU 11.4.0
C compiler flags: -fexcess-precision=fast -funroll-all-loops -mavx2 -mfma -Wno-missing-field-initializers -O3 -DNDEBUG
C++ compiler: /usr/bin/c++ GNU 11.4.0
C++ compiler flags: -fexcess-precision=fast -funroll-all-loops -mavx2 -mfma -Wno-missing-field-initializers -Wno-cast-function-type-strict -fopenmp -O3 -DNDEBUG
BLAS library: Internal
LAPACK library: Internal

Running on 1 node with total 8 cores, 16 processing units
Hardware detected on host Zen-2:
CPU info:
  Vendor: AMD
  Brand: AMD Ryzen 7 4800H with Radeon Graphics
  Family: 23 Model: 96 Stepping: 1
  Features: aes amd apic avx avx2 clflush cmov cx8 cx16 f16c fma htt lahf misalignsse mmx msr nonstop_tsc pclmuldq pdep1gb popcnt pse rdrnd rdtsop sha sse2 sse3 sse4a sse4.1 sse4.2 sse4.3
Hardware topology: Basic
Packages, cores, and logical processors:
[indices refer to OS logical processors]
  Package 0: [ 0 1] [ 2 3] [ 4 5] [ 6 7] [ 8 9] [ 10 11] [ 12 13] [ 14 15]
CPU limit set by OS: -1 Recommended max number of threads: 16
```

	Core t (s)	Wall t (s)	(%)
Time:	200.599	12.546	1598.9
	(ns/day)	(hour/ns)	
Performance:	688.660	0.035	

```
Started mdrun on rank 0 Mon Oct 5 10:48:06 2026

Step      Time
0         0.00000

Energies (kJ/mol)
  Angle      LJ (SR)  Disper. corr.  Coulomb (SR)  Coul. recip.
  6.08769e+02  1.40611e+03  -7.50233e+01  -9.59162e+03  2.05056e+02
  Potential    Kinetic En.  Total Energy  Conserved En.  Temperature
-7.44671e+03  1.68554e+03  -5.76118e+03  -5.75405e+03  2.90227e+02
Pres. DC (bar) Pressure (bar) Constr. rmsd
-1.56065e+02  -1.35361e+03  2.82196e-06

Step      Time
5000     10.00000

Energies (kJ/mol)
  Angle      LJ (SR)  Disper. corr.  Coulomb (SR)  Coul. recip.
  5.21556e+02  1.77912e+03  -9.13966e+01  -1.15498e+04  6.46798e+01
  Potential    Kinetic En.  Total Energy  Conserved En.  Temperature
-9.27587e+03  1.82024e+03  -7.45563e+03  -5.42601e+03  3.13421e+02
Pres. DC (bar) Pressure (bar) Constr. rmsd
-2.31437e+02  -2.53113e+03  3.06580e-06

Step      Time
10000    20.00000

Energies (kJ/mol)
  Angle      LJ (SR)  Disper. corr.  Coulomb (SR)  Coul. recip.
  5.35164e+02  1.96661e+03  -1.03401e+02  -1.17647e+04  6.35793e+01
  Potential    Kinetic En.  Total Energy  Conserved En.  Temperature
-9.30279e+03  1.86570e+03  -7.43709e+03  -5.24068e+03  3.21248e+02
Pres. DC (bar) Pressure (bar) Constr. rmsd
-2.96103e+02  5.99902e+02  3.05472e-06
```

MOLECULAR MODELING: OUTPUT FILES

ener.edr: Properties history of simulation

Let's check the temperature of the system:

- `gmx energy -f ener.edr -o temperature.xvg`
- Select «9 Temperature» and double «Enter»

In the end you can see

- **Average value of the property**
- **Estimated error of the property**
- **Root mean square deviation** (standard deviation)
- **Total drift** (how much the property has changed from the start to the end of the simulation, indicating how well equilibrated the system is)

- `xmgrace temperature.xvg`

```
Command line:
  gmx energy -f ener.edr -o temperature.xvg

Opened ener.edr as single precision energy file

Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
-----
  1  LJ-(SR)           2  Disper.-corr.      3  Coulomb-(SR)       4  Coul.-recip.
  5  Potential         6  Kinetic-En.       7  Total-Energy      8  Conserved-En.
  9  Temperature      10  Pres.-DC         11  Pressure          12  Box-X
 13  Box-Y            14  Box-Z           15  Volume            16  Density
 17  pV               18  Enthalpy        19  Vir-XX            20  Vir-XY
 21  Vir-XZ           22  Vir-YX          23  Vir-YY            24  Vir-YZ
 25  Vir-ZX           26  Vir-ZY          27  Vir-ZZ            28  Pres-XX
 29  Pres-XY          30  Pres-XZ          31  Pres-YX            32  Pres-YY
 33  Pres-YZ          34  Pres-ZX          35  Pres-ZY            36  Pres-ZZ
 37  #Surf*SurfTen    38  T-System        39  Lamb-System

9

Back Off! I just backed up temperature.xvg to ./#temperature.xvg.1#
Last energy frame read 10 time 100.000

Statistics over 50001 steps [ 0.0000 through 100.0000 ps ], 1 data sets
All statistics are over 501 points

Energy                Average  Err.Est.      RMSD  Tot-Drift
-----
Temperature           297.582    0.63    12.2333    2.25651 (K)
```

MOLECULAR MODELING: OUTPUT FILES

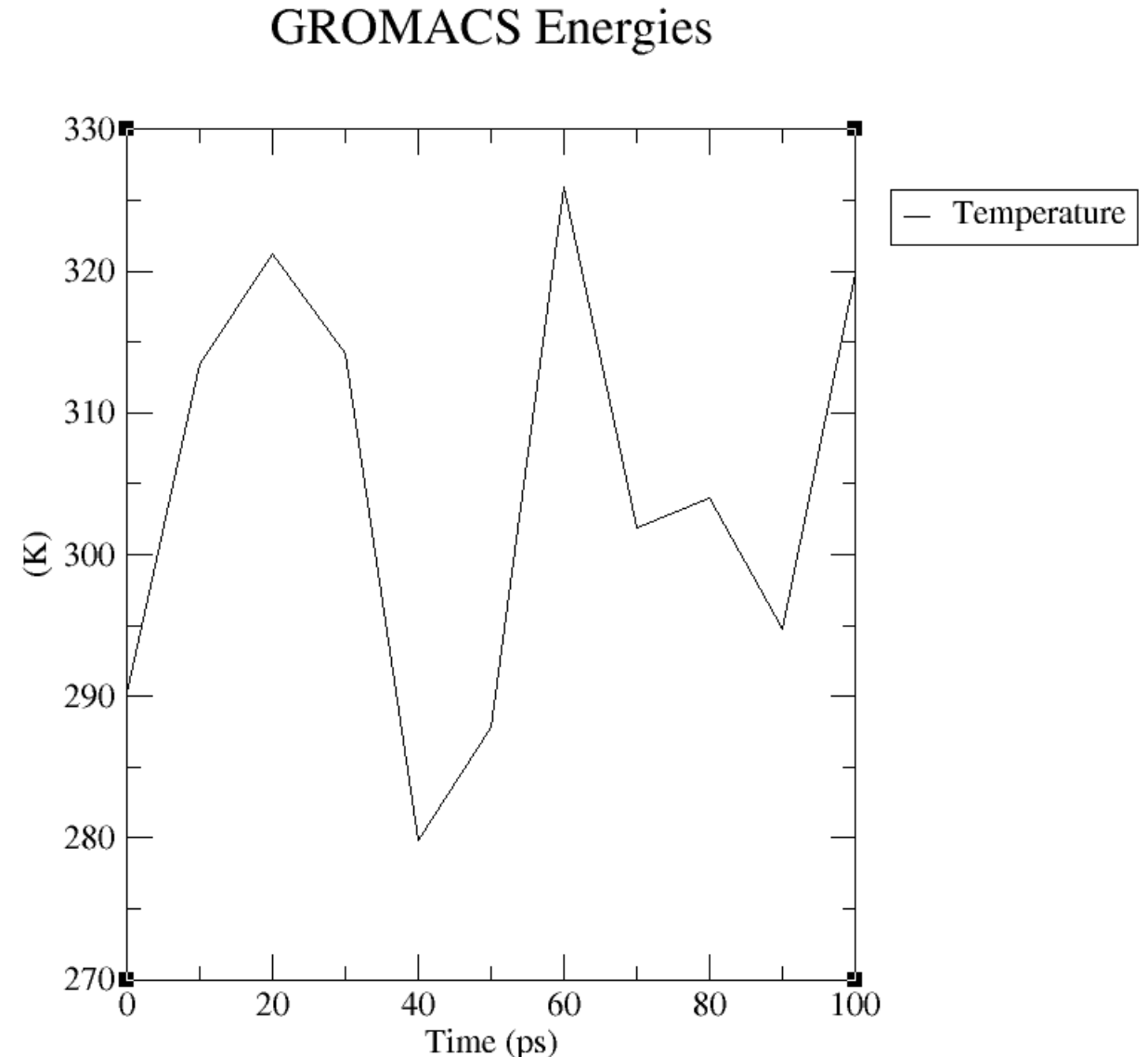
ener.edr: Properties history of simulation

Let's check the temperature of the system:

- `gmx energy -f ener.edr -o temperature.xvg`
- Select «9 Temperature» and double «Enter»

In the end you can see

- **Average value of the property**
 - **Estimated error of the property**
 - **Root mean square deviation** (standard deviation)
 - **Total drift** (how much the property has changed from the start to the end of the simulation, indicating how well equilibrated the system is)
- `xmgrace temperature.xvg`



MOLECULAR MODELING: SIMULATION QUALITY CHECK

Let's check 4 more properties of the system:

- **Density**
- Radial Distribution Function
- Diffusion coefficient
- Enthalpy of vaporization

Density

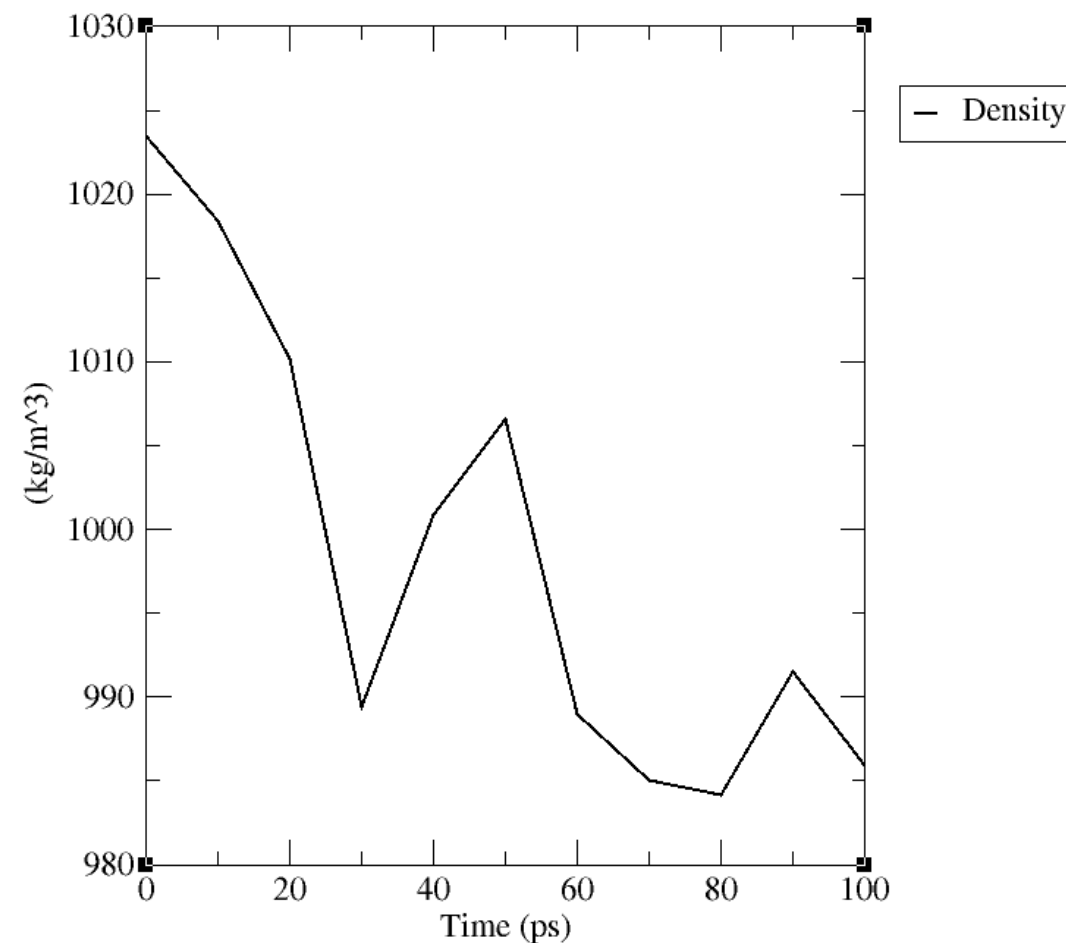
- `gmx energy -f ener.edr -o density.xvg`
- Select «16 Density» and double «Enter»

Energy	Average	Err.Est.	RMSD	Tot-Drift	
Density	1001.41	3.4	16.7051	-23.0026	(kg/m ³)

Close to target

Must be small numbers

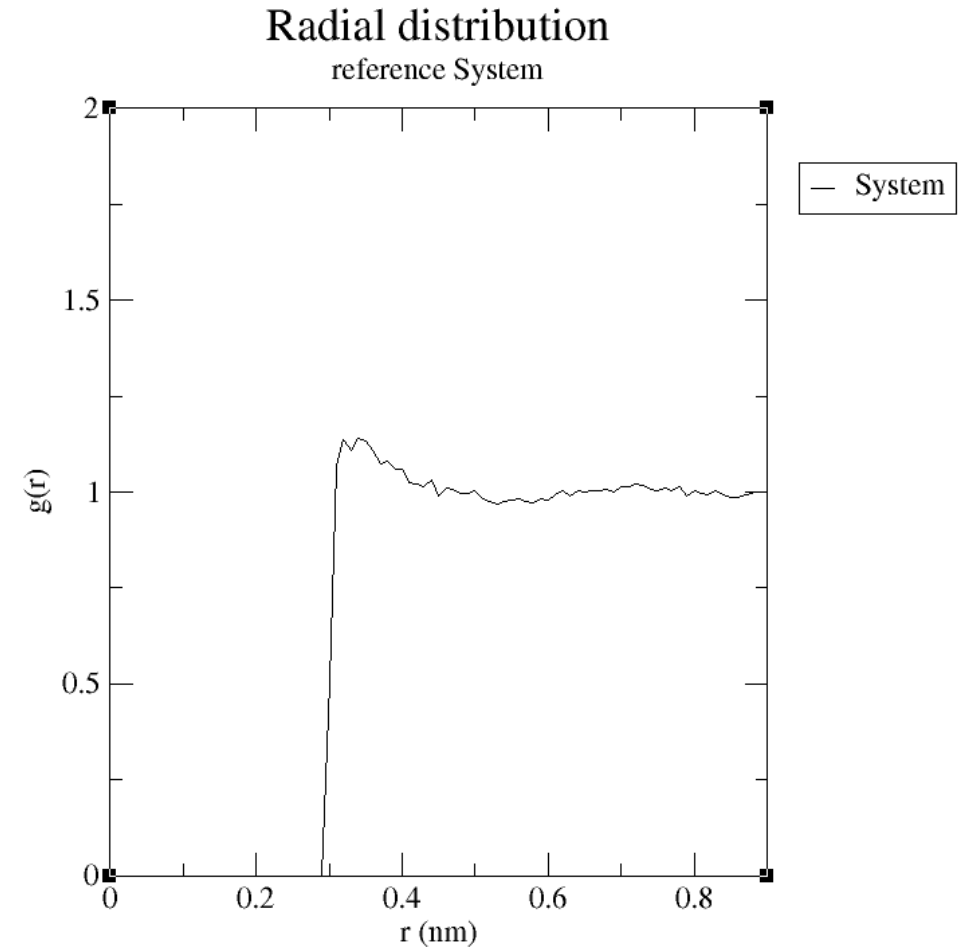
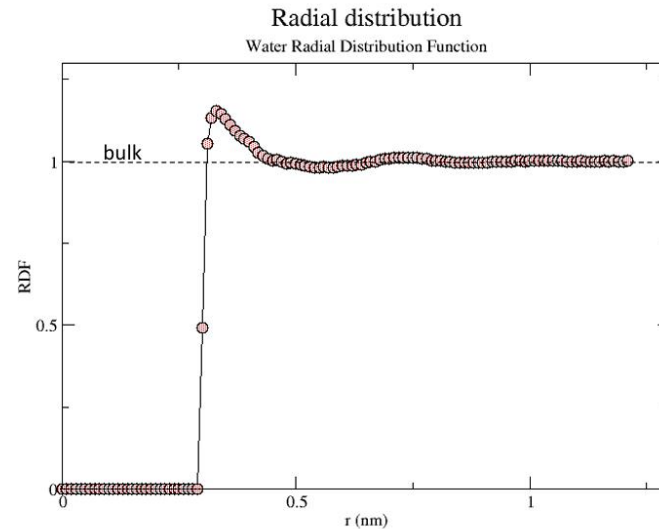
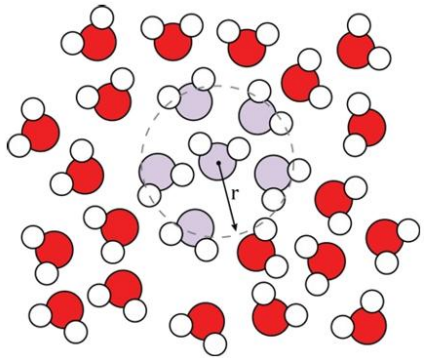
GROMACS Energies



MOLECULAR MODELING: SIMULATION QUALITY CHECK

Let's check 4 more properties of the system:

- Density
- **Radial Distribution Function**
- Diffusion coefficient
- Enthalpy of vaporization



Radial Distribution function

- `gmx rdf -f traj.trr -s simul.tpr -cut 0.3 -bin 0.01 -o rdf.xvg`
- Select «Group 0 (System)» for both reference and selection, then CTRL+D

MOLECULAR MODELING: SIMULATION QUALITY CHECK

Let's check 4 more properties of the system:

- Density
- Radial Distribution Function
- Diffusion coefficient
- Enthalpy of vaporization

Diffusion coefficient

$$\vec{J} = -\mathbf{D} \vec{\nabla} \Phi$$

Fick's first law

Let's calculate the *self-diffusion coefficient* (D) by using the Einstein eq. (1905):

$$D = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \langle \Delta \mathbf{r}(t)^2 \rangle$$

$\langle \Delta \mathbf{r}(t)^2 \rangle = \langle |\mathbf{r}_1(t) - \mathbf{r}_1(0)|^2 \rangle$ is defined as mean square displacement

$$\langle \Delta \mathbf{r}(t)^2 \rangle = \frac{1}{N} \sum_{i=1}^N |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2$$

Note 1: We time average the distance covered in a time t for each i -th particle.

Note 2: We are able to compute a macroscopic properties D from a microscopic quantity $\langle \Delta \mathbf{r}(t)^2 \rangle$

NB: Here we use another trajectory file since the previous one is too short (100 ps). This one is 20 ns.

- `gmx msd -f traj20ns.trr -s simul.tpr -type x -o msd_x.xvg -trestart 100`
- Select «Group 0 (System)», then CTRL+D
- `xmgrace msd_x.xvg >` identify the linear region in the log-log plot (es. 5000-15000 ps)
- `gmx msd -f traj20ns.trr -s simul.tpr -type x -o msd_x_fit.xvg -trestart 100 -beginfit 5000 -endfit 15000`
- Fitting also for Y and Z

MOLECULAR MODELING: SIMULATION QUALITY CHECK

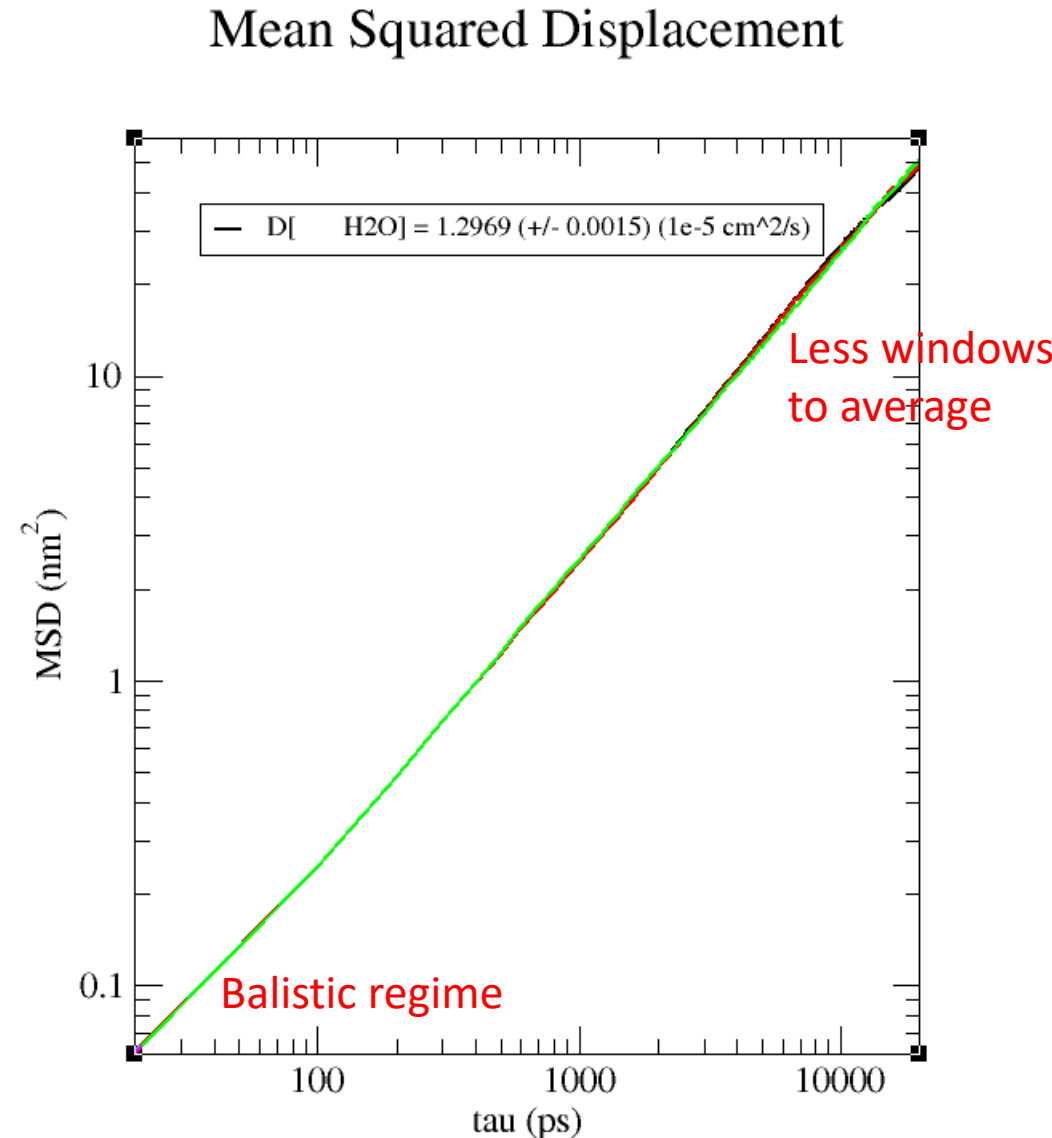
Let's check 4 more properties of the system:

- Density
- Radial Distribution Function
- **Diffusion coefficient**
- Enthalpy of vaporization

Diffusion coefficient

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MOLECULAR MODELING: SIMULATION QUALITY CHECK

Let's check 4 more properties of the system:

- Density
- Radial Distribution Function
- Diffusion coefficient
- Enthalpy of vaporization

Enthalpy of vaporization ΔH_{vap} is the amount of energy needed to transform a liquid in a gas.

- $\Delta H_{\text{vap}} = \Delta U_{LG} + p\Delta V$ is the variation of internal energy U (gas phase) – U (liquid phase) plus $p\Delta V$ (work) against external pressure.
- ΔU_{LG} represents the energy needed to break intermolecular interactions in the liquid phase.

Again, we can compute a macroscopic properties (ΔH_{vap}) from microscopic quantities.

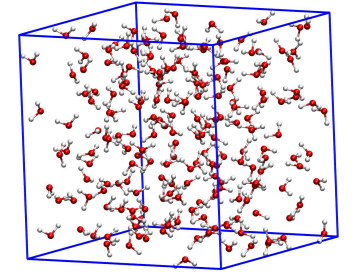
$$\Delta H_{\text{vap}} = \langle U_{\text{vap}} \rangle - \frac{\langle U_{\text{liq}} \rangle}{N_{\text{mol}}} + RT$$

R gas constant, T is the temperature (K), $\langle U_{\text{vap}} \rangle$ e $\langle U_{\text{liq}} \rangle$ average internal energies of H_2O in the vapor and liquid phases.

MOLECULAR MODELING: SIMULATION QUALITY CHECK

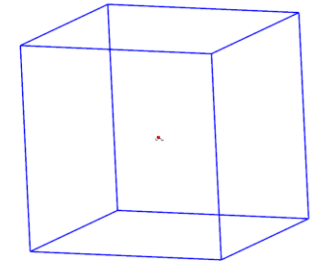
Let's check 4 more properties of the system:

- Density
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- Diffusion coefficient
- Enthalpy of vaporization



You need a simulation with one molecule in vacuum (GAS) and a simulation of liquid bulk (LIQ).

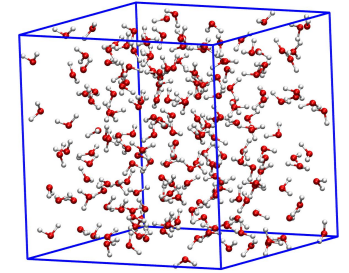
- mkdir singlemolecule
- cp **topol.top** **spce.itp** **mdprod.mdp** **one_water.gro** singlemolecule
- cd singlemolecule and correct number of molecule in **topol.top**
- gmx grompp -f mdprod.mdp -p topol.top -c one_water.gro -o simul.tpr
- gmx mdrun -s simul.tpr
- gmx energy -f **ener.edr** -o **enthalpy.xvg**
- Select «1-LJ (SR)» «2-Disper. Corr.» «3-Coulomb (SR)» «4-Coul. Recip.» and double «Enter» (GAS)
- cd ../ to go back in liquid box directory
- gmx energy -f **ener.edr** -o **enthalpy.xvg**
- Select «1-LJ (SR)» «2-Disper. Corr.» «3-Coulomb (SR)» «4-Coul. Recip.» and double «Enter» (LIQUID)



MOLECULAR MODELING: SIMULATION QUALITY CHECK

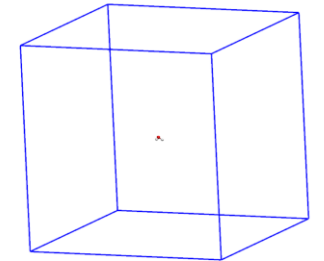
Let's check 4 more properties of the system:

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- Diffusion coefficient
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- gmx energy -f **ener.edr** -o **enthalpy.xvg**
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- cd ../ to go back in liquid box directory
- gmx energy -f **ener.edr** -o **enthalpy.xvg**



Energy	Average	Err.Est.	RMSD	Tot-Drift	
LJ (SR)	1923.71	12	109.944	-70.3079	(kJ/mol)
Disper. corr.	-143.012	0.49	2.39121	3.29265	(kJ/mol)
Coulomb (SR)	-11267.6	20	169.456	130.686	(kJ/mol)
Coul. recip.	112.482	0.51	10.2974	2.8776	(kJ/mol)

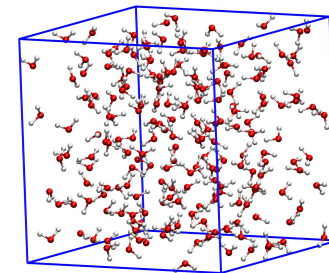
Energy	Average	Err.Est.	RMSD	Tot-Drift	
LJ (SR)	0	0	0	0	(kJ/mol)
Disper. corr.	0	0	0	0	(kJ/mol)
Coulomb (SR)	-8.1158	2.4e-06	3.72879e-05	-1.30709e-05	(kJ/mol)
Coul. recip.	7.86516	0.0027	0.0538257	0.0172773	(kJ/mol)



MOLECULAR MODELING: SIMULATION QUALITY CHECK

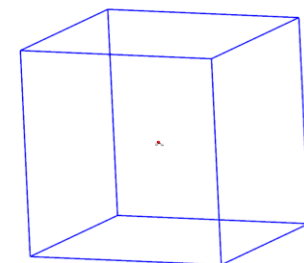
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- cd singlemolecule and correct number of molecule in **topol.top**
- gmx grompp -f mdprod.mdp -p topol.top -c one_water.gro -o simul.tpr
- gmx mdrun -s simul.tpr
- gmx energy -f **ener.edr** -o **enthalpy.xvg**
- Select «1-LJ (SR)» «2-Disper. Corr.» «3-Coulomb (SR)» «4-Coul. Recip.» and double «Enter» (GAS)
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49.1
[kJ/mol]